

Answer: C

19 Using First Law of Thermodynamics, $\Delta U = Q + W$

For process PRQ,
 $\Delta U = 8 - 3 = 5 \text{ J}$

For process PSQ,
 $\Delta U = Q - 1 = 5 \text{ J}$
 $Q = 6 \text{ J}$

Answer: B

20 At sea level, T is approximately 300 K